

Testing serial independence of object-valued time series

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SUMMARY

We propose a novel method for testing serial independence of object-valued time series in metric spaces, which are more general than Euclidean or Hilbert spaces. The proposed method is fully nonparametric, free of tuning parameters and can capture all nonlinear pairwise dependence. The key concept used in this paper is the distance covariance in metric spaces, which is extended to the autodistance covariance for object-valued time series. Furthermore, we propose a generalized spectral density function to account for pairwise dependence at all lags and construct a Cramér–von Mises-type test statistic. New theoretical arguments are developed to establish the asymptotic behaviour of the test statistic. A wild bootstrap is also introduced to obtain the critical values of the nonpivotal limiting null distribution. Extensive numerical simulations and two real data applications on cumulative intraday returns and human mortality data are conducted to illustrate the effectiveness and versatility of our proposed test.

Some key words: Distance covariance; Non-Euclidean-valued data; Random object; Spectral test; White noise testing.

1. INTRODUCTION

Random objects in general metric spaces have become increasingly common in modern statistical and econometric research. For example, the intraday return path of a financial asset (Aue et al., 2017), the annual composition of energy sources (Zhu & Müller, 2023b), social networks (Board & Meyer-ter Vehn, 2021), EEG scans or MRI fiber tracts of patients (Kurtek et al., 2012) can all be viewed as random objects in certain metric spaces, although they are typically given specific names such as functional data, compositional data, network data, image data and curve data, among others. The concept of random objects also undoubtedly includes classical notions such as vectors, covariance matrices and distributions. Instead of building specific models for each of them, by viewing these objects as random elements in metric spaces, we may be able to simplify the modelling and inference while preserving the ability of extracting meaningful information and patterns in a unified fashion (Dubey & Müller, 2019, 2020; Petersen & Müller, 2019; Zhang et al., 2022).

For many endeavours in this area, the data they analysed are collected with a natural ordering, i.e., the data can be viewed as an object-valued time series. However, most existing modelling and inference techniques either presume temporal independence or construct time series models without conducting diagnostic checks to assess the goodness of fit. This is mainly due to the unavailability of appropriate tests. As a result, researchers often overlook the serial dependence in their data and fail to account for its impact on their analyses. Consequently, the validity and reliability of their findings could be compromised. This motivates us to develop a new test for serial independence of object-valued time series.

Testing serial independence has a long and rich history in statistics and econometrics, with a vast literature that cannot be exhaustively listed. The early work dates back to [Box & Pierce \(1970\)](#) and [Ljung & Box \(1978\)](#). Since then, numerous tests have been proposed for univariate random variables or multivariate vectors in Euclidean space, with much attention devoted to testing for second-order uncorrelatedness; see [Li & McLeod \(1981\)](#), [Deo \(2000\)](#), [Lobato \(2001\)](#), [Escanciano & Lobato \(2009\)](#), [Shao \(2011\)](#), to name a few. These tests typically capture linear serial dependence in data, and have no power against nonlinear dependence. Nonlinear serial dependence is indeed prevalent among many real-world time series, and many parametric nonlinear models have been proposed to capture nonlinear dependence. A prominent example is the generalized autoregressive conditional heteroskedasticity model, which implies uncorrelatedness, but is serially dependent. Several tests have been developed to target higher-order dependence ([Li & Mak, 1994](#); [Ling & Li, 1997](#)) and at general nonlinear dependence ([Hong, 1999](#); [Escanciano & Velasco, 2006](#)). Developments in the testing of the white noise hypothesis in a functional time series in Hilbert space have been discussed by, e.g., [Gabrys & Kokoszka \(2007\)](#), [Horváth et al. \(2013\)](#) and [Zhang \(2016\)](#).

Despite many tests being available for testing the serial independence or uncorrelatedness in Euclidean and Hilbert spaces, they cannot be directly used for testing serial independence of object-valued time series, because of the lack of classic algebraic operations in general metric spaces, such as addition, multiplication and taking the inner product. To fill this gap, we propose a new test based on the distance covariance, which was originally proposed by [Székely et al. \(2007\)](#) to measure dependence among two random vectors in Euclidean space using characteristic functions, and later extended by [Zhou \(2012\)](#) to time series settings by using the autodistance covariance, ADCV. A crucial feature of the (auto)distance covariance is that it can capture both linear and nonlinear serial dependencies in data. Although many researchers have employed this idea in serial dependence testing, it is only valid in the Euclidean space ([Fokianos & Pitsillou, 2017, 2018](#); [Davis et al., 2018](#)).

Based on the concept of ‘distance covariance in metric space’ by [Lyons \(2013\)](#), we are able to extend ADCV in Euclidean space to general metric spaces, which then naturally serves the goal of testing independence at fixed lags. To take into account pairwise (in)dependence at all lags, we then propose a generalized spectral density/distribution function using ADCVs in the same spirit as a classical spectral density/distribution function using autocovariances. Following the developments of [Shao \(2011\)](#), who proposed a spectrum-based test for the white noise hypothesis of a univariate time series, we develop a Cramér–von Mises type test statistic and study the limiting behaviour of the test under both the null and alternatives. Our test significantly enhances classical spectrum-based tests in two fundamental ways. First, it captures both linear and nonlinear serial dependencies, a capability lacking in conventional methods except for those of [Hong \(1999\)](#) and [Escanciano & Velasco \(2006\)](#). Second, its versatility extends to data objects in general metric spaces, which are much broader than the Euclidean or Hilbert spaces considered in the literature. To the best of our knowledge,

this is the first formal attempt at testing temporal independence for object-valued time series in a unified manner. Unlike conventional autocovariance function or spectrum-based tests for Euclidean time series, the estimation of ADCVs is based on U-statistics. Therefore, new theoretical arguments are developed to establish the asymptotic theory for the proposed test statistic. Since the limiting null distribution is nonstandard and nonpivotal, we propose a wild bootstrap approach to facilitate practical implementation of our test. The bootstrap consistency is also established.

We now introduce some notation. Let (Ω, d) be a separable metric space, and let $(\mathbb{S}, \mathbb{P}, \mathcal{F})$ be the probability space. For a vector $a \in \mathbb{R}^q$, we denote its Euclidean norm as $|a|_q$. Let $\xrightarrow{\mathbb{P}}$ and $\rightarrow$ denote the convergence in probability and in distribution, respectively. We denote by $L_2[0, \pi]$ the Hilbert space $\mathbb{H}$ of all square integrable functions on $[0, \pi]$, with respect to the Lebesgue measure, with inner product $\langle f, g \rangle = \int_{[0, \pi]} f(\zeta) g^c(\zeta) d(\zeta)$, where $g^c(\zeta)$ denotes the complex conjugate of $g(\zeta)$, and the norm $\|f\| = \langle f, f \rangle^{1/2}$. By $\Rightarrow$ we denote weak convergence in $L_2[0, \pi]$.

2. PRELIMINARIES

2.1. Distance covariance

The concept of ‘distance covariance’ was first introduced by Székely et al. (2007) as a measure of dependence between two random vectors $X \in \mathbb{R}^p$ and $Y \in \mathbb{R}^q$. It is defined as the weighted integral of the discrepancy between the joint characteristic function and the product of the marginal characteristic functions of (X, Y) .

DEFINITION 1 (DISTANCE COVARIANCE IN EUCLIDEAN SPACE). For $X \in \mathbb{R}^p$ and $Y \in \mathbb{R}^q$, the distance covariance is

$$\begin{aligned} \text{dcov}(X, Y) = & \int_{\mathbb{R}^{p+q}} |\mathbb{E}[\exp\{i(t'X + s'Y)\}] - \mathbb{E}[\exp(it'X)]\mathbb{E}[\exp(is'Y)]|^2 \\ & \times (c_p c_q |t|_p^{1+p} |s|_q^{1+q})^{-1} dt ds, \end{aligned}$$

where $c_d = \pi^{(1+d)/2} / \Gamma\{(1+d)/2\}$ and Γ is the gamma function.

For notational simplicity, we use dcov instead of dcov^2 as used in the original definition of distance covariance. Székely et al. (2007) also managed to derive the following alternative definition when $\mathbb{E}[|X|_p^2 + |Y|_q^2] < \infty$:

$$\begin{aligned} \text{dcov}(X, Y) = & \mathbb{E}|X - X'|_p |Y - Y'|_q + \mathbb{E}|X - X'|_p \mathbb{E}|Y - Y''|_q \\ & - 2\mathbb{E}|X - X'|_p |Y - Y''|_q \end{aligned} \quad (1)$$

with (X', Y') and (X'', Y'') independent copies of (X, Y) .

Distance covariance can be used to characterize the dependence between X and Y due to the following crucial property.

PROPOSITION 1. Distance covariance is nonnegative if $\text{dcov}(X, Y) \geq 0$; the equality holds if and only if X is independent of Y .

Proposition 1 has proven to be a very powerful tool in Euclidean space and has been used in mutual dependence testing (Yao et al., 2018), feature screening (Li et al., 2012), dimension

reduction (Sheng & Yin, 2016), among many other applications. However, Definition 1 is not easily extended to general metric spaces because conventional algebraic manipulation, such as addition and multiplication, may not be applied. Instead, by viewing $|\cdot|_q$ as a metric in Euclidean space, the alternative definition in (1) is extendable.

2.2. Distance covariance in metric space

Let (Ω, d) be a separable metric space. Let $M(\Omega)$ be the set of probability measures on Ω ; we denote the subset of $M(\Omega)$ possessing p th moment as

$$M_p(\Omega) = \left\{ \nu \in M(\Omega) : \text{for some } \omega \in \Omega, \int_{\Omega} d^p(\omega, x) d\nu(x) < \infty \right\}.$$

For $\nu \in M_1(\Omega)$, define (Lyons, 2013, Lemma 2.1)

$$\begin{aligned} d^{(1)}(x) &= \int_{\Omega} d(x, x') d\nu(x'), & D &= \int_{\Omega} d^{(1)}(x) d\nu(x), \\ d_{\nu}(x, x') &= d(x, x') - d^{(1)}(x) - d^{(1)}(x') + D. \end{aligned}$$

Clearly, for two independent random objects $X, X' \sim \nu(\cdot)$, we can write $d^{(1)}(X) = \mathbb{E}\{d(X, X') | X\}$ and $D = \mathbb{E}\{d(X, X')\}$.

If X and Y are two random vectors taking values in conventional q -dimensional Euclidean space, i.e., $(\Omega, d) = (\mathbb{R}^q, |\cdot|_q)$, with marginal distributions ν_X and ν_Y , respectively, then (1) is equivalent to $\text{dcov}(X, Y) = \mathbb{E}\{d_{\nu_X}(X, X')d_{\nu_Y}(Y, Y')\}$. Therefore, Lyons (2013) proposed the following definition of distance covariance in general metric spaces.

DEFINITION 2 (DISTANCE COVARIANCE IN METRIC SPACE). For X, Y taking values in (Ω, d) whose marginals are $\nu_X, \nu_Y \in M_1(\Omega)$, respectively, the distance covariance between X and Y is

$$\text{dcov}(X, Y) = \mathbb{E}\{d_{\nu_X}(X, X')d_{\nu_Y}(Y, Y')\},$$

where (X', Y') is an independent copy of (X, Y) .

However, as pointed out by Lyons (2013), in general metric spaces, the above definition alone is insufficient for Proposition 1 to hold, and additional topological assumptions are required. The concept of *strong negative type* was introduced by Lyons (2013) to resolve this issue.

DEFINITION 3 (STRONG NEGATIVE TYPE). We say that (Ω, d) is of strong negative type if, for $\nu_1, \nu_2 \in M_1(\Omega)$ such that $\nu_1 \neq \nu_2$, and $\nu_- = \nu_1 - \nu_2$, we have

$$\iint_{\Omega^2} d(x_1, x_2) d\nu_-(x_1) d\nu_-(x_2) < 0.$$

The class of metric spaces of strong negative type is actually quite large; for example, every separable Hilbert space is of strong negative type. However, there are many spaces that do not satisfy Definition 3; for example, $\mathbb{R}^q$ with L_p metric for $3 \leq q \leq \infty$, $2 < p \leq \infty$ is not of strong negative type. We refer the reader to Lyons (2013, 2014, 2020) for further discussion and examples. With the notion of strong negative type, distance covariance

then completely characterizes the (in)dependence in the metric space, given in the following proposition.

PROPOSITION 2 (THEOREM 3.11 OF LYONS, 2013). *If (Ω, d) is of strong negative type, for X, Y taking values in (Ω, d) whose marginals are $\nu_X, \nu_Y \in M_1(\Omega)$, respectively, Proposition 1 continues to hold.*

2.3. Autodistance covariance

The distance covariance, as defined in Definition 2, measures the dependence between two random objects and might not be readily applicable to the time series context. Therefore, in order to address this limitation, we introduce the concept of autodistance covariance. This notion was first proposed by Zhou (2012) for conventional Euclidean-valued stationary time series. First, we follow Rio (2017) to extend the concept of strict stationarity to metric spaces, which implies that the joint distributions of the process are invariant under translations of time.

DEFINITION 4 (STRICT STATIONARITY; DEFINITION 1.1 OF RIO, 2017). *The time series sequence $\{X_t\}_{t \in \mathbb{Z}}$ in (Ω, d) is strictly stationary if, for any positive integer t and any subset $\mathbb{S}$ of $\mathbb{Z}$,*

$$\{X_{s+t} : s \in \mathbb{S}\} \text{ has the same distribution as } \{X_s : s \in \mathbb{S}\}.$$

In Zhou (2012), ADCV is defined by measuring temporal dependence between $\{X_t\}_{t \in \mathbb{Z}} \in \mathbb{R}^q$ and its lagged observation $\{X_{t-k}\}_{t \in \mathbb{Z}}$ for a fixed lag order $k \in \mathbb{Z}$. Here, we generalize the idea of Zhou (2012) to time series objects in metric space.

DEFINITION 5 (ADCV). *Assume that $\{X_t\}_{t \in \mathbb{Z}}$ is a sequence of strict stationary time series taking values in (Ω, d) . For $k \in \mathbb{Z}$, we call*

$$V(k) = \mathbb{E}\{d_v(X_t, X'_t)d_v(X_{t-k}, X'_{t-k})\}$$

the autodistance covariance of X_t at lag k , where $\{X'_t\}_{t \in \mathbb{Z}}$ is an independent copy of $\{X_t\}_{t \in \mathbb{Z}}$.

It is clear that $V(k) = V(-k)$ for $k < 0$ and $V(0) = \mathbb{E}\{d_v^2(X_t, X'_t)\} > 0$. In addition, by Proposition 2, $V(k) = 0$, $k \neq 0$, if and only if X_t and X_{t-k} are independent of each other. We exploit this property to build our test statistics in the next section.

Remark 1. In Zhou (2012), the ADCV for the Euclidean-valued time series is based on Definition 1 by replacing (X, Y) with (X_t, X_{t-k}) . However, for general metric spaces, such treatment is no longer valid. We thus call for Definition 2. It can be shown that, for $(\Omega, d) = (\mathbb{R}^q, |\cdot|_q)$, Definition 5 is equivalent to that in Zhou (2012).

3. TEST STATISTICS AND ASYMPTOTICS

3.1. Test statistics at fixed lags

Given a sequence of stationary random objects $\{X_t\}_{t=1}^n$ that reside in a separable metric space (Ω, d) of strong negative type, we are interested in testing the serial independence of

$\{X_t\}_{t=1}^n$. To illustrate the idea, we first consider a relatively simple task by forming a test based on ADCV at fixed lags $1 \leq k \leq K$, i.e., we are interested in

$$\begin{aligned} H_0: \{X_t\}_{t=1}^n & \text{ is independent and identically distributed} \\ \text{versus } H_a: V(k) & \neq 0 \quad \text{for some } 1 \leq k \leq K. \end{aligned}$$

This suggests that we find an empirical estimator for $V(k)$. Intuitively, under H_0 , $(X_{t'}, X_{t'-k})$ naturally forms an independent copy of (X_t, X_{t-k}) if $t' \neq t$, which motivates us to estimate $V(k)$ by replacing terms involving (conditional) expectations in $d_v(X_t, X_{t-k})$ with their empirical counterparts.

Motivated by the estimator in Euclidean space (Székely et al., 2007), we construct the empirical estimator $V_n(k)$ by adopting the $\mathcal{U}$ -centring approach of Székely & Rizzo (2014). Specifically, let $a_{ij}(k) = d(X_i, X_j)$ for $k+1 \leq i, j \leq n$; by $\{\tilde{a}_{ij}(k)\}_{i,j=k+1,\dots,n}$ we denote its $\mathcal{U}$ -centred version:

$$\tilde{a}_{ij}(k) = \begin{cases} a_{ij} - \frac{\sum_{t=k+1}^n a_{it}}{n-k-2} - \frac{\sum_{t'=k+1}^n a_{t'j}}{n-k-2} + \frac{\sum_{k+1 \leq t \neq t' \leq n} a_{tt'}}{(n-k-1)(n-k-2)}, & i \neq j, \\ 0, & i = j. \end{cases} \quad (2)$$

Define $\tilde{b}_{ij}(k)$ similarly for $b_{ij}(k) = d(X_{i-k}, X_{j-k})$ with $k+1 \leq i, j \leq n$. Intuitively, for large n , $\tilde{a}_{ij}(k)$, or $\tilde{b}_{ij}(k)$, approximates the value of $d_v(X_i, X_j)$ or $d_v(X_{i-k}, X_{j-k})$.

We then estimate $V(k)$ by

$$V_n(k) = \frac{1}{(n-k)(n-k-3)} \sum_{i,j=k+1}^n \tilde{a}_{ij}(k) \tilde{b}_{ij}(k).$$

The $\mathcal{U}$ -centring approach was originally proposed by Székely & Rizzo (2014) to provide an unbiased estimator of the distance covariance. It is adopted here to simplify the technical analysis as we can alternatively rewrite $V_n(k)$ using the fourth order U-statistic (Zhang et al., 2018)

$$V_n(k) = \binom{n-k}{4}^{-1} \sum_{k+1 \leq i < j < q < r \leq n} h(Z_i^{(k)}, Z_j^{(k)}, Z_q^{(k)}, Z_r^{(k)}), \quad (3)$$

where $Z_i^{(k)} = (X_i, X_{i-k})$ and the kernel function is

$$h(Z_i, Z_j, Z_q, Z_r) = \frac{1}{24} \sum_{(i_1, i_2, i_3, i_4)}^{(i,j,q,r)} d(X_{i_1}, X_{i_2}) \{d(Y_{i_3}, Y_{i_4}) + d(Y_{i_1}, Y_{i_2}) - 2d(Y_{i_1}, Y_{i_3})\},$$

where $Z_\ell = (X_\ell, Y_\ell)$, $\ell \in \{i, j, q, r\}$, and $\sum_{(i_1, i_2, i_3, i_4)}^{(i,j,q,r)}$ denotes the summation over all $24 (= 4!)$ permutations of the 4-tuple of indices (i, j, q, r) .

Remark 2. Alternatively, one could use the conventional empirical estimator of $V(k)$ given by

$$\hat{V}_n(k) = \frac{1}{(n-k)^2} \sum_{i,j=k+1}^n \hat{a}_{ij}(k) \hat{b}_{ij}(k),$$

where

$$\begin{aligned} \hat{a}_{ij}(k) &= a_{ij}(k) - \bar{a}_{i.}(k) - \bar{a}_{.j}(k) + \bar{a}_{..}(k) \\ \bar{a}_{i.}(k) &= \frac{1}{n-k} \sum_{t=k+1}^n a_{it}, \quad \bar{a}_{.j}(k) = \frac{1}{n-k} \sum_{t=k+1}^n a_{tj}, \quad \bar{a}_{..}(k) = \frac{1}{(n-k)^2} \sum_{t,t'=k+1}^n a_{t,t'}, \end{aligned}$$

with $\hat{b}_{ij}(k)$ defined similarly. As $\hat{V}_n(k)$ is a combination of V-statistics, the technical analysis could be more involved than the $\mathcal{U}$ -centering approach (Székely & Rizzo, 2014).

To analyse the asymptotic behaviours of $V_n(k)$, we make the following assumption.

Assumption 1. The marginal distribution $\nu(\cdot) \in M_4(\Omega)$.

The following theorem derives the limiting distribution of (3) by exploiting the properties of U-statistics; see, e.g., Lee (1990). We emphasize here that, even under H_0 , $V_n(k)$ is not a conventional U-statistic with kernels applied to independent and identically distributed samples. For example, $Z_t^{(k)} = (X_t, X_{t-k})$ and $Z_{t+k}^{(k)} = (X_{t+k}, X_t)$ are not independent of each other. Therefore, more delicate technical treatments are required.

THEOREM 1. Under H_0 , suppose that Assumption 1 holds. Then, fix $K \geq 1$; as $n \rightarrow \infty$,

$$\{(n-k)V_n(k)\}_{k=1}^K \xrightarrow{d} \{\xi_k\}_{k=1}^K,$$

where $\xi_k \stackrel{d}{=} \sum_{\ell=1}^{\infty} \lambda_{\ell} [\{G_{\ell}^{(k)}\}^2 - 1]$. Here $\{G_{\ell}^{(k)}\}_{\ell=1,2,\dots,\infty; k=1,\dots,K}$ is a sequence of independent and identically distributed standard Gaussian random variables, and $\{\lambda_{\ell}\}_{\ell=1}^{\infty}$ and $\{e_{\ell}(\cdot)\}_{\ell=1}^{\infty}$ are sequences of nonzero eigenvalues and orthonormal eigenfunctions corresponding to

$$\lambda_{\ell} e_{\ell}(z) = \mathbb{E}\{d_{\nu}(x, X) d_{\nu}(y, Y)\} e_{\ell}(z),$$

where $z = (x, y) \in \Omega^2$ and $Z = (X, Y) \in \Omega^2$ with X and Y independent copies of X_t .

Theorem 1 improves the result of Zhou (2012) under H_0 , not only by extending the result in Euclidean space to more general metric spaces, but also by providing joint convergence of sample ADCVs. In addition, Theorem 1 is crucial for proving the asymptotics of the spectrum-based test below. In practice, $\{\xi_k\}_{k=1}^K$ is a sequence of the mixture of independently and identically distributed centred $\chi^2(1)$ random variables, which are nonpivotal. Below we adopt a wild bootstrap method to approximate the limiting null distributions; we defer the details to §4.

3.2. Generalized spectral test

We then consider the hypothesis involving all lag orders, i.e.,

$$\begin{aligned} H_0: \{X_t\}_{t=1}^n &\text{ is independent and identically distributed} \\ \text{versus } H_a: V(k) &\neq 0 \quad \text{for some } k \neq 0. \end{aligned}$$

For our testing purpose, we propose the generalized spectral density

$$f(\zeta) = (2\pi)^{-1} \sum_{k=-\infty}^{\infty} V(k)e^{-ik\zeta}, \quad \zeta \in [-\pi, \pi], \quad (4)$$

and the generalized spectral distribution function $F(\zeta) = \int_0^\zeta f(z) dz$, $\zeta \in [0, \pi]$. Clearly, $f(\cdot)$, or $F(\cdot)$, is motivated by the spectral density, or distribution, function in Euclidean space and Hilbert space, where the classical spectral density function is defined by replacing $V(k)$ in (4) by $\gamma(k)$, with $\gamma(k) = \text{cov}(X_t, X_{t-k})$ the autocovariance at lag k .

Our generalized spectral density, or function, serves as a generalization of the classical spectral density, or function, in two significant aspects. First, we aim to test serial independence, which is a stronger notion than serial uncorrelatedness. In the conventional spectral density function, $\gamma(k)$ is the autocovariance measuring linear correlation between X_t and its lag- k observation X_{t-k} , which implies that spectrum-based tests can only capture linear serial dependence in the second-order structure. By contrast, if we consider $V(k)$, the autodistance covariance at lag k , we can additionally measure the lag- k nonlinear dependence. Second, the definition of autocovariance requires certain algebraic operations, such as subtraction and the inner product in the conventional Euclidean or Hilbert space, whereas the autodistance covariance does not, making it applicable to more general metric spaces.

Under H_0 , because $V(k) = 0$ for all $k \neq 0$, it holds that $F(\zeta) = V(0)\Psi_0(\zeta)$; on the other hand, under H_a , $F(\zeta) = \sum_{k=0}^{\infty} V(k)\Psi_k(\zeta)$, where $\Psi_k(\zeta) = \sin(k\zeta)/(k\pi)\mathbb{1}(k \neq 0) + \zeta/(2\pi)\mathbb{1}(k = 0)$. This observation implies that we can construct a test by comparing the empirical estimator of $F(\zeta) - V(0)\Psi_0(\zeta)$ and zero. In particular, we define the process

$$S_n(\zeta) = \sum_{k=1}^{n-4} (n-k)V_n(k)\Psi_k(\zeta), \quad \zeta \in [0, \pi], \quad (5)$$

and consider the Cramér–von Mises-type statistic

$$\text{CvM}_n = \int_0^\pi S_n^2(\zeta) d\zeta.$$

In (5), the summation ends with $k = n - 4$ because a minimum of four observations are required to compute $V_n(k)$ effectively. Additionally, the choice of normalizing constant as $n - k$ instead of n is made to enhance finite sample performance.

Similar Cramér–von Mises-type statistics have been adopted in Euclidean space and Hilbert space for testing second-order white noise or martingale differences; see, e.g., [Deo \(2000\)](#), [Escanciano & Velasco \(2006\)](#), [Shao \(2011\)](#) and [Zhang \(2016\)](#), among many others.

Remark 3. One could also consider Kolmogorov–Smirnov-type statistics of the form

$$\text{KS}_n = \sup_{\zeta \in [0, \pi]} |S_n(\zeta)|. \quad (6)$$

However, the sup functional is no longer a continuous map in L_2 Hilbert space, and therefore the process convergence result below and the continuous mapping theorem do not lead to the asymptotic null distribution of the Kolmogorov–Smirnov test statistic, and a rigorous theoretical investigation is beyond the scope of this paper. Nevertheless, our numerical studies in §5 below suggest that KS_n also delivers satisfactory performance.

Remark 4. We briefly discuss the difference between our tests and the distance covariance-based tests by [Fokianos & Pitsillou \(2017, 2018\)](#) in Euclidean space.

First, their test is designed for Euclidean-valued time series, while ours has much more broad applications in other metric spaces. Second, their test is founded on the basis of [Hong \(1999\)](#) by noting the connection between the characteristic function-based definition of distance covariance (cf. Definition 1) and the dependence metric defined by [Hong \(1999\)](#) in terms of the weighting function. However, their approach is no longer valid in the metric space. Third, their test statistic is based on a kernel weighted sum of empirical ADCVs at a set of lags. Therefore, they require an additional tuning parameter to truncate the kernel, which is typically known to affect the convergence rate and hence power. On the contrary, our test is tuning-free, and is more powerful even in the Euclidean setting, as demonstrated in the [Supplementary Material](#).

Our next theorem obtains the weak convergence result for $S_n(\zeta)$ in the sense of the L_2 metric.

THEOREM 2. *Under H_0 , suppose that Assumption 1 holds. Then*

$$\{S_n(\zeta)\}_{\zeta \in [0, \pi]} \Rightarrow \{S(\zeta)\}_{\zeta \in [0, \pi]} \quad \text{in } L_2[0, \pi],$$

where $\{S(\zeta)\}_{\zeta \in [0, \pi]} \stackrel{d}{=} \{\sum_{k=1}^{\infty} \xi_k \Psi_k(\zeta)\}_{\zeta \in [0, \pi]}$ with $\{\xi_k\}$ defined in Theorem 1.

Based on Theorem 2 and because the integral functional is continuous, the following corollary is a natural consequence of the continuous mapping theorem.

COROLLARY 1. *Under H_0 , suppose that Assumption 1 holds. Then*

$$\text{CvM}_n \xrightarrow{d} \int_0^\pi S^2(\zeta) d\zeta.$$

Therefore, given significance level $\alpha \in (0, 1)$, we reject H_0 if $\text{CvM}_n \geq \text{CvM}(1 - \alpha)$, where $\text{CvM}(1 - \alpha)$ denotes the $1 - \alpha$ quantile of $\int_0^\pi S^2(\zeta) d\zeta$. For practical implementation, we need to invoke the bootstrap method in §4 to approximate the above critical value.

To study the behaviours of the tests under H_a , we impose the following β -mixing conditions on $\{X_t\}_{t \in \mathbb{Z}}$. Basically, it requires weak temporal dependence of the process, with similar assumptions adopted by [Fokianos & Pitsillou \(2017, 2018\)](#) and [Davis et al. \(2018\)](#) for distance covariance-based inference in time series. Weak temporal dependence is also required in [Zhou \(2012\)](#) using physical dependence measures ([Wu, 2005](#)).

Assumption 2. It holds that $\{X_t\}_{t=1}^n$ is a stationary β -mixing process, with marginal distribution $\nu(\cdot) \in M_{2+2\delta}(\Omega)$ for some $\delta > 0$. Furthermore, the mixing coefficient satisfies that $\beta(n) = O(n^{-r})$ for some $r > 0$.

Theorem 3 below derives the asymptotics of the test statistics under H_a .

THEOREM 3. *Suppose that Assumption 2 holds. We have $V_n(k) \xrightarrow{P} V(k)$ for $k = 1, 2, \dots$. Under H_a , as $n \rightarrow \infty$, $n^{-1} \text{CvM}_n \xrightarrow{P} \sum_{k=1}^{\infty} V^2(k) \int_0^\pi \Psi_k^2(\zeta) d\zeta$.*

Clearly, under H_a , we have $V(k) > 0$ for some $k \neq 0$, which implies that our test has asymptotically power 1 under H_a .

4. BOOTSTRAP

In this section, we use the wild bootstrap method to approximate the limiting null distribution of CvM_n , and show its asymptotic validity.

Algorithm 1. Let $\mathbf{w} = (w_1, \dots, w_n)$ be a sequence of independent random variables following the Rademacher distribution: $\mathbb{P}(w_i = 1) = \mathbb{P}(w_i = -1) = \frac{1}{2}$.

1. For each $k = 1, \dots, n-4$, define the doubly weighted bootstrapped version of V_n by

$$V_n^*(k)^{(b)} = \frac{1}{(n-k)(n-k-3)} \sum_{k+1 \leq i \neq j \leq n} w_i^{(b)}(k) \tilde{a}_{ij} \tilde{b}_{ij} w_j^{(b)}(k), \quad (7)$$

where $\tilde{a}_{ij}$ and $\tilde{b}_{ij}$ are $\mathcal{U}$ -centring versions of $a_{ij} = d(X_i, X_j)$ and $b_{ij} = d(X_{i-k}, X_{j-k})$, respectively, and $\mathbb{w}^{(b)}(k)$ is an independent random realization of $\mathbb{w}$ with $w_i^{(b)}(k)$ the i th element of $\mathbb{w}^{(b)}(k)$.

2. Obtain the bootstrapped process

$$S_n^*(\zeta) = \sum_{k=1}^{n-4} (n-k) V_n^*(k)^{(b)} \Psi_k(\zeta), \quad \zeta \in [0, \pi],$$

and compute $\text{CvM}_n^{*(b)} = \int_0^\pi [S_n^*(\zeta)]^2 d\zeta$.

3. Repeat steps 1–2 B times and collect $\{\text{CvM}_n^{*(b)}\}_{b=1}^B$.
4. Denote the α th upper percentile of $\{\text{CvM}_n^{*(b)}\}_{b=1}^B$ by $\text{CvM}_n^*(1-\alpha)$, and reject H_0 if $\text{CvM}_n > \text{CvM}_n^*(1-\alpha)$.

For each fixed k , the bootstrapped autodistance covariance estimator $V_n^*(k)^{(b)}$ can be viewed as a realization of the independent wild bootstrap for degenerate U-statistics; see [Dehling & Mikosch \(1994\)](#) and [Lee et al. \(2020\)](#). They are further combined to obtain the bootstrap realization of $\text{CvM}_n^{*(b)}$. Here we use independent copies of wild bootstrap weights $\mathbb{w}(k)$ for different k values. This is implied by Theorem 1 that despite common components in constructing $\{V_n(k)\}_{k=1}^K$, their asymptotic distributions $\{\xi_k\}_{k=1}^K$ are independent of each other under H_0 .

Remark 5. For our test targeting at the global null of serial independence, independent wild bootstrap is sufficient. If one is interested in making statistical inference for $V_n(k)$ alone, potential serial dependence at other lag orders should be accounted for. A possible extension along these lines would be to pinpoint the minimum lag at which dependencies exist by sequentially testing the nullity of $V_n(k)$. To achieve this, the above wild bootstrap method should be adjusted accordingly. For example, [Leucht & Neumann \(2013\)](#) used the dependent wild bootstrap of [Shao \(2010\)](#) by constructing temporarily dependent weights $\{w_i\}_{i=1}^n$, as an extension of the wild bootstrap of [Dehling & Mikosch \(1994\)](#) for degenerate U-statistics. See also [Zhou \(2012\)](#) for a subsampling-based procedure.

Remark 6. Another approach to approximate the limiting null distribution is to use permutation. Although both bootstrap and permutation methods can be based on pre-calculated pairwise distance metrics $d(X_i, X_j)$ for $1 \leq i < j \leq n$, permutation is less

computationally efficient. This is because in the $\mathcal{U}$ -centring step, the permutation method needs to recalculate $\tilde{a}_{ij}$ (or $\tilde{b}_{ij}$) in (2), which involves the summation of $d(X_{\tau(i)}, X_{\tau(j)})$ with $\tau(i)$ the permutation order of the i th original observation. In contrast, the bootstrap method does not require such recalculation, because $\tilde{a}_{ij}$ (or $\tilde{b}_{ij}$) is kept constant, as shown in (7), across all bootstrap statistics.

We then proceed to analyse the asymptotic behaviour of the bootstrapped test statistics. We first introduce the notion of convergence in distribution in probability; see Definition 2 of Li et al. (2003).

DEFINITION 6 (CONVERGENCE IN DISTRIBUTION IN PROBABILITY). *For a sequence of bootstrapped statistics T_n^* that depends on the random sample $\{X_t\}_{t=1}^n$, we say that $T_n^* \mid (X_1, X_2, \dots)$ converges to $T \mid (X_1, X_2, \dots)$ in distribution in probability if, for any subsequence T'_n , there exists a further subsequence T''_n such that $T''_n \mid (X_1, X_2, \dots)$ converges to $T \mid (X_1, X_2, \dots)$ in distribution for almost every sequence $(X_1, X_2, \dots)$. We use the notation*

$$T_n^* \xrightarrow{d^*} T \quad \text{in probability.}$$

THEOREM 4. *Under H_0 , suppose that Assumption 1 holds. For $\mathbb{W}$ such that $\mathbb{E}w_i = 0$, $\mathbb{E}w_i^2 = 1$ and $\mathbb{E}w_i^4 < \infty$. Then, for any fixed $K \geq 1$, as $n \rightarrow \infty$,*

$$\{(n-k)V_n^*(k)\}_{k=1}^K \xrightarrow{d^*} \{\xi_k\}_{k=1}^K \quad \text{in probability,}$$

where ξ_k is defined in Theorem 1.

THEOREM 5. *Under the same conditions as in Theorem 4,*

$$\text{CvM}_n^* \xrightarrow{d^*} \int_0^\pi S^2(\zeta) d\zeta \quad \text{in probability,}$$

where $\{S(\zeta)\}_{\zeta \in [0, \pi]} \stackrel{d}{=} \{\sum_{k=1}^\infty \xi_k \Psi_k(\zeta)\}_{\zeta \in [0, \pi]}$ with $\{\xi_k\}$ defined in Theorem 1. Furthermore, we have $\mathbb{P}\{\text{CvM}_n > \text{CvM}_n^*(1 - \alpha)\} \rightarrow \alpha$.

Therefore, the wild bootstrap in Algorithm 1 provides a consistent approximation of the limiting null distribution of our test statistics. When H_0 is violated, we show that the bootstrapped tests have consistent powers.

THEOREM 6. *Suppose that Assumption 2 holds with $\delta = 1$; then, under H_a , for $\alpha \in (0, 1)$, we have*

$$\mathbb{P}\{\text{CvM}_n > \text{CvM}_n^*(1 - \alpha)\} \rightarrow 1.$$

5. SIMULATION STUDIES

5.1. Functional time series

We first consider the simulation studies for functional time series, where the underlying space is $\Omega = \{f: [0, 1] \mapsto \mathbb{R}, \int_0^1 f^2(\tau) d\tau < \infty\}$ with L_2 metric, i.e., $d(f, g) = [\int_0^1 \{f(t) - g(t)\}^2 dt]^{1/2}$ for any $f, g \in \Omega$.

Let $\mathcal{B} := \{B(\tau): \tau \in [0, 1]\}$ denote the standard Brownian motion. We mimic the setups of Zhang (2016) to generate $\{Y_t(\tau): \tau \in [0, 1]\}_{t=1}^n$ from the following data-generating

processes, corresponding to Brownian motion (BM), the Brownian bridge (BB), the functional autoregressive conditional heteroscedasticity (FARCH) model, the functional non-linear moving average (FNMA) model, and the functional autoregressive (FAR) model, respectively.

- (i) BM: $Y_t(\tau) = \varepsilon_t(\tau)$ for $\tau \in [0, 1]$, where $\{\varepsilon_t(\tau)\}_{t=1}^n \stackrel{\text{i.i.d.}}{\sim} \mathcal{B}$.
- (ii) BB: $Y_t(\tau) = \varepsilon_t(\tau) - \tau \varepsilon_t(1)$ for $\tau \in [0, 1]$, where $\{\varepsilon_t(\tau)\}_{t=1}^n \stackrel{\text{i.i.d.}}{\sim} \mathcal{B}$.
- (iii) FARCH model: $Y_t(\tau_1) = \varepsilon_t(\tau_1)[\tau_1 + \int_0^1 \rho \exp\{(\tau_1^2 + \tau_2^2)/2\} Y_{t-1}^2(\tau_2) d\tau_2]^{1/2}$ for $\tau_1 \in [0, 1]$, where $\{\varepsilon_t(\tau_1)\}_{t=1}^n \stackrel{\text{i.i.d.}}{\sim} \mathcal{B}$. We consider $\rho \in \{0.3418, 0.6153\}$, which corresponds to the cases when the Hilbert–Schmidt norm of the Gaussian kernel equals 0.5 and 0.9, respectively.
- (iv) FNMA model: $Y_t(\tau) = \varepsilon_t(\tau)\varepsilon_{t-1}(\tau)$ for $\tau \in [0, 1]$, where $\{\varepsilon_t(\tau)\}_{t=1}^n$ is either BM or BB.
- (v) FAR model: $Y_t(\tau_1) = \int_0^1 \mathcal{K}(\tau_1, \tau_2) Y_{t-1}(\tau_2) d\tau_2 + \varepsilon_t(\tau_1)$ for $\tau_1 \in [0, 1]$, where $\{\varepsilon_t(\tau_1)\}_{t=1}^n$ is generated by either BM or BB, and we consider the Gaussian kernel $\mathcal{K}_G(\tau_1, \tau_2) = c_G \exp\{(\tau_1^2 + \tau_2^2)/2\}$ and the Wiener kernel $\mathcal{K}_W(\tau_1, \tau_2) = c_W \min\{\tau_1, \tau_2\}$. For each kernel, we choose the coefficient c so that the corresponding Hilbert–Schmidt norm equals 0.3, i.e., $c_G = 0.2051$ for $\mathcal{K}_G$ and $c_W = 0.7346$ for $\mathcal{K}_W$.

In addition to the wild bootstrap method, denoted `BooT`, introduced in §4, here and in §§ 5.2 and 5.3 below, we also apply the standard permutation test, denoted `Permt`, to approximate the critical value for the proposed Cramér–von Mises test. Furthermore, we outline the results of the Kolmogorov–Smirnov test in (6) using the same bootstrapped or permuted samples of $S_n^*(\cdot)$. Both bootstrap and permutation conduct $R = 300$ replicates. Among the existing tests that target serial dependence within a sequence of functional data, we also consider the following proposed by Gabrys & Kokoszka (2007), Horváth et al. (2013) and Zhang (2016) for comparison.

- (i) Gabrys & Kokoszka (2007), denoted GK, proposed the portmanteau test based on functional principal component analysis. We consider the lag truncation number $h \in \{1, 3, 5\}$ and select the smallest P such that the cumulative variation explained by the first P principal components is above 90%.
- (ii) Horváth et al. (2013), denoted HHR, presented an independence test based on the empirical correlation functions. We aggregate the recommendations of Horváth et al. (2013) and Zhang (2016) to consider the lag truncation number $h \in \{5, 10, 30, 50\}$. Each integral involved in HHR is approximated by the Riemann sum with $p = 100$ points.
- (iii) Zhang (2016) proposed a Cramér–von Mises-type test, denoted Zhang, based on the functional periodogram. The critical value of Zhang is approximated using block bootstraps with $R = 300$ replicates, and we consider the block size $b \in \{1, 5, 10\}$ as well as the optimal size b^* chosen by the minimal volatility method. All the integrals are approximated by the Riemann sum with $p = 50$ points to ease the computational burden.

We fix $n = 200$ and generate the data on a grid of $T = 1000$ equally spaced points in $[0, 1]$ for each functional observation. We use Fourier and B -splines, order 4, with $B = 20$ basis functions to obtain the functional data.

Table 1 includes the empirical rejection rate under each data-generating process over 2500 Monte Carlo replicates. Overall, the type of basis functions has limited influence on the finite sample performance for all the methods in Table 1. Under the Brownian motion and

Table 1. Empirical rejection rates (%) of the functional time series when $n = 200$

Data-generating process		Proposed				GK			HHHR				Zhang			
		CvM-P	CvM-B	KS-P	KS-B	$h = 1$	$h = 3$	$h = 5$	$h = 5$	$h = 10$	$h = 30$	$h = 50$	$b = 1$	$b = 5$	$b = 10$	b^*
BM	F	5.9	5.7	5.9	6.0	4.8	5.2	4.0	6.8	6.4	6.4	7.1	5.3	5.7	6.2	6.2
	B	5.5	5.8	5.7	6.0	4.7	5.0	4.1	6.8	6.4	6.3	7.1	5.5	5.4	6.0	6.2
BB	F	5.2	5.1	5.0	5.0	4.0	4.4	3.4	5.7	5.4	6.8	7.2	4.3	3.8	3.6	4.3
	B	5.1	5.4	4.7	5.1	3.9	4.2	3.2	5.7	5.4	6.8	7.2	4.6	3.8	3.7	4.4
FARCH(0.5)	F	11.3	8.5	11.9	8.1	12.7	11.1	8.7	10.6	9.9	8.6	8.6	5.2	4.6	5.1	5.6
	B	11.4	9.0	11.4	8.1	12.7	11.1	8.7	10.6	10.0	8.6	8.7	4.7	4.3	5.0	5.3
FARCH(0.9)	F	29.6	19.4	28.2	17.3	28.8	27.9	24.0	21.3	16.4	10.7	9.0	4.5	3.7	4.1	4.5
	B	29.4	18.4	27.5	17.4	28.7	27.8	24.1	21.3	16.3	10.8	9.0	4.3	3.7	3.8	4.6
FNMA-BM	F	83.9	62.6	81.0	58.5	37.3	22.2	17.2	18.0	14.9	10.2	8.4	3.9	3.9	4.5	4.6
	B	84.1	62.4	80.4	59.1	37.2	22.5	17.1	17.8	14.8	10.1	8.4	4.2	4.0	4.5	4.5
FNMA-BB	F	77.4	51.8	73.0	49.6	51.2	26.9	18.8	21.1	16.4	11.3	11.0	4.4	2.8	2.6	3.5
	B	77.4	51.8	73.0	49.4	49.2	26.2	18.5	20.8	16.4	11.0	10.8	4.2	2.7	2.8	3.3
FAR-G-BM	F	98.4	98.4	98.1	98.1	96.4	83.2	72.6	92.1	82.2	66.2	55.9	97.3	97.0	95.8	97.3
	B	98.5	98.4	98.1	98.0	96.1	82.8	72.8	92.1	81.8	65.6	55.7	97.2	96.8	95.7	97.2
FAR-G-BB	F	98.4	98.5	97.8	98.1	97.2	82.7	69.2	86.6	74.1	54.0	45.3	95.7	94.2	92.4	94.4
	B	98.5	98.3	97.9	98.0	96.6	81.8	68.2	86.6	74.1	54.1	45.4	96.0	93.8	93.2	94.3
FAR-W-BM	F	97.3	97.2	97.1	97.0	91.9	78.6	67.8	91.8	86.4	71.8	63.6	98.4	97.9	96.8	98.3
	B	97.4	97.3	96.6	97.1	92.0	78.8	68.0	92.0	86.5	71.9	63.5	98.4	97.4	97.5	98.0
FAR-W-BB	F	96.8	96.9	96.6	96.4	93.3	70.6	55.0	85.1	71.5	52.0	44.1	95.0	91.4	90.5	93.1
	B	97.4	97.1	96.7	97.1	93.6	71.2	56.2	85.7	71.9	52.1	44.4	95.3	93.1	91.3	93.9

BM, Brownian motion; BB, Brownian bridge; FARCH, functional autoregressive conditional heteroscedasticity model; FNMA, functional nonlinear moving average model; FAR, functional autoregressive model; P, Permt; B, Boot; F, the Fourier basis; B, B -spline; CvM, Cramér–von Mises; KS, Kolmogorov–Smirnov.

Brownian bridge null models, the empirical size of the Cramér–von Mises and Kolmogorov–Smirnov tests are both close to 0.05, and the wild bootstrap has a similar size accuracy to the permutation test. In contrast, the size accuracies of GK and HHHR depend on the lag truncation parameter h , and moderate size distortion can be observed with a larger number of lags. Zhang achieves decent size accuracy with a smaller block size b , but has some size distortion with a large b .

As for the power behaviours, our tests, especially with Permt, demonstrate dominating performance in the FNMA and FAR models. While they are slightly inferior in FARCH(0.5), the performances get better when the dependence level increases in FARCH(0.9). In comparison, both GK and HHHR seem to be sensitive to truncation parameter selections, leading to a considerable loss in power, especially with a larger lag truncation parameter. Both the FARCH and FNMA models have serial dependence, but no serial correlation, and thus correspond to the null models of Zhang (2016). Consequently, the power of Zhang against these models is close to the nominal level. GK and Zhang can have comparable power with our tests, but only when a smaller bandwidth, or block size, parameter is used. Overall, our tests are among the best across all settings.

5.2. Covariance matrix

Next, we consider the scenario where the observations are a sequence of covariance matrices $\{Y_t\}_{t=1}^n \in \mathbb{R}^{p \times p}$, which is relatively new in the literature.

We generate the time series from a conditional autoregressive Wishart model (Golosnoy et al., 2012) with dimension $p \in \{2, 5, 8\}$. In particular, let $\mathcal{W}_p(d, V)$ denote the Wishart distribution with degrees of freedom d and scale matrix V , and we generate $\{\varepsilon_t\}_{t=1}^n \stackrel{\text{i.i.d.}}{\sim} \mathcal{W}_p(10, I_p) \in \mathbb{R}^{p \times p}$. Furthermore, we generate $Y_t = \frac{1}{10} \text{chol}(\Sigma_t) \varepsilon_t \text{chol}(\Sigma_t)$, where $\text{chol}(\Sigma_t)$ denotes the lower-triangular Cholesky factor of Σ_t , and $\Sigma_t = CC^T + \rho AY_{t-1}A^T + \rho B \Sigma_{t-1} B^T$ with $A = 0.7I_p$, $B = 0.5I_p$ and $C = I_p$. Here we use the parameter $\rho \in \{0, 0.5, 75\}$ to control the temporal dependence and it is trivial that $\rho = 0$ corresponds to the null model. We use $\text{CAW}_p(\rho)$ to denote the model with dimension p and parameter ρ .

In this subsection, we have $\Omega = \{A: \text{symmetric, positive definite matrix} \in \mathbb{R}^{p \times p}\}$. To evaluate the distance between two covariance matrices $A, B \in \Omega$, we consider the following metrics.

- (i) Euclidean metric: $d(A, B) = \|A - B\|_F$.
- (ii) Log-Euclidean metric: $d(A, B) = \|\log m(A) - \log m(B)\|_F$, where $\log m(\cdot)$ denotes the logarithm of a matrix.
- (iii) Cholesky metric: $d(A, B) = \|\text{chol}(A) - \text{chol}(B)\|_F$, where $\text{chol}(A)$ denotes the Cholesky decomposition of A .
- (iv) Riemann metric: $d(A, B) = \|\log m(A^{-1/2}BA^{-1/2})\|_F$, where $A^{1/2}$ denotes the matrix square root of A .

In view of the fact that there is little literature regarding this testing problem, we use the half vectorization to convert a $p \times p$ covariance matrix into a $\frac{1}{2}p(p+1)$ -dimensional vector and apply the multivariate testing methods to the resulting time series. In addition to our proposed test with the Euclidean metric, denoted Vech-Euc , we also adopt the test statistic mADCv proposed by Fokianos & Pitsillou (2018), which can be implemented via the R package dCovTS (R Development Core Team, 2024). In particular, mADCv is based on the autodistance covariance matrix and requires a kernel function $k(\cdot)$ and a bandwidth parameter p_n . We follow the set-up of Fokianos & Pitsillou (2018) to consider the truncated kernel, the Daniell kernel, the Parzen kernel and the Bartlett kernel. Here, we set $p_n = \lfloor 3n^\lambda \rfloor$ with $\lambda \in \{1/10, 2/10, 3/10\}$. Based on 1000 Monte Carlo replicates, all the numerical results are tabulated in Table 2.

According to Table 2, our test achieves accurate size, regardless of the dimension p and the choice of metric. Under the alternative, our proposed test with the Euclidean metric demonstrates the highest power when applied to either matrix-valued time series or half-vectorized multivariate time series. Interestingly, the power increases as dimension p grows, indicating its advantage in the moderate-dimensional scenario. On the other hand, the log-Euclidean metric and the Riemann metric in our tests exhibit high power when p is small, but they do show a slight power loss against moderate values of p . We conjecture that this discrepancy in performance could be attributed to the fact that the log-Euclidean metric and Riemann metric are immune to scaling changes. In the case of alternatives like the CAW model, which seeks to model dynamic movements across all elements of the covariance matrices, metrics such as Euc and Vech-Euc , which consider changes in all elements, are likely to demonstrate greater efficacy and power. In general, different metrics excel at capturing distinct dependence structures in covariance matrices; we refer the interested reader to § 5 of Dryden et al. (2009) for comparisons and discussions.

In contrast, mADCv has a conservative size under the null and has noticeable power loss under the alternative, especially when p is moderate and ρ is small. When $p = 5$ and $p = 8$, the half-vectorized covariance matrix is converted to the multivariate time series in $\mathbb{R}^{15}$ and $\mathbb{R}^{36}$, respectively. The facts that mADCv involves the estimation of the long-run covariance matrix and the estimation tends to be less accurate as the dimension grows may explain why mADCv worsens against growing p .

5.3. Univariate distribution

Lastly, we consider the case where Ω is the set of cumulative distribution functions of a random variable that takes values from $[0, 1]$. In this case, we mimic the settings of Zhu & Müller (2023a) to generate a sequence of cumulative distribution functions $\{Y_t(x): x \in [0, 1]\}_{t=1}^n$ with $n = 200$ from an autoregressive transport model of order p , denoted $\text{ATM}(p)$.

Table 2. Empirical rejection rates (%) of the covariance matrix time series generated by the CAW_p(ρ) model

Method			$p = 2$			$p = 5$			$p = 8$		
			$\rho = 0$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 0$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 0$	$\rho = 0.5$	$\rho = 0.75$
Proposed	Euc	CvM-P	4.3	41.9	99.7	6.7	57.0	100.0	4.2	74.5	100.0
		CvM-B	4.4	41.2	99.9	6.9	55.8	100.0	4.6	74.4	100.0
		KS-P	4.5	40.3	99.7	6.1	52.9	100.0	4.6	68.1	100.0
		KS-B	4.8	39.4	99.7	7.2	51.7	100.0	5.3	69.4	100.0
	log-Euc	CvM-P	5.1	34.7	99.7	5.9	30.9	100.0	6.1	25.6	100.0
		CvM-B	4.9	34.9	99.6	5.8	31.6	100.0	6.1	25.4	100.0
		KS-P	4.8	34.0	99.3	6.0	29.2	100.0	6.3	22.6	100.0
		KS-B	5.1	33.9	99.4	6.5	29.4	100.0	6.8	22.3	100.0
	Chol	CvM-P	5.2	38.7	99.5	4.5	34.0	100.0	5.9	34.8	100.0
		CvM-B	5.2	37.8	99.6	4.5	33.9	100.0	6.5	35.6	100.0
		KS-P	4.6	35.5	99.5	4.9	31.4	100.0	6.1	30.5	100.0
		KS-B	5.0	35.6	99.5	5.1	32.1	100.0	6.5	31.5	100.0
	Riemann	CvM-P	5.3	34.6	99.6	6.2	31.6	100.0	5.8	29.0	100.0
		CvM-B	5.3	34.5	99.5	6.1	31.4	100.0	6.2	28.7	100.0
		KS-P	5.1	34.2	99.5	6.8	29.3	100.0	5.8	26.2	100.0
		KS-B	5.3	34.0	99.4	6.8	29.8	100.0	7.0	25.9	100.0
	Vech-Euc	CvM-P	4.8	39.9	99.4	5.9	52.7	100.0	4.4	70.3	100.0
		CvM-B	4.9	38.9	99.4	6.2	51.7	100.0	4.2	69.7	100.0
		KS-P	5.2	37.6	99.2	5.8	50.0	100.0	4.6	63.5	100.0
		KS-B	4.9	37.4	99.3	6.1	49.1	100.0	4.7	63.9	100.0
mADCV	TC	$\lambda = 1/10$	3.4	14.6	89.9	0.0	0.0	89.8	0.0	0.0	66.6
		$\lambda = 2/10$	3.6	13.1	87.1	0.0	0.1	93.1	0.0	0.0	89.5
		$\lambda = 3/10$	3.7	11.9	82.2	0.0	0.3	96.3	0.0	0.0	98.3
	DAN	$\lambda = 1/10$	2.9	25.3	98.4	0.0	0.2	98.0	0.0	0.0	84.3
		$\lambda = 2/10$	2.9	20.9	97.1	0.0	0.3	97.3	0.0	0.0	91.5
		$\lambda = 3/10$	3.4	17.9	93.9	0.0	0.3	97.9	0.0	0.0	97.4
	PAR	$\lambda = 1/10$	2.6	24.3	98.3	0.0	0.3	97.7	0.0	0.0	83.7
		$\lambda = 2/10$	2.6	19.0	96.4	0.0	0.2	96.9	0.0	0.0	88.5
		$\lambda = 3/10$	3.7	17.5	93.2	0.0	0.3	97.7	0.0	0.0	97.4
	BAR	$\lambda = 1/10$	2.6	25.7	98.5	0.0	0.5	97.9	0.0	0.0	74.4
		$\lambda = 2/10$	2.6	21.3	97.6	0.0	0.4	98.1	0.0	0.0	89.0
		$\lambda = 3/10$	3.0	20.0	95.8	0.0	0.3	98.1	0.0	0.0	95.3

mADCV, test proposed by Fokianos & Pitsillou (2018); Euc, Euclidian metric; Chol, Cholesky metric; Riemann, Riemann metric; Vech-Euc, our proposed test with the Euclidean metric; TC, truncated kernel; DAN, Daniell kernel; PAR, Parzen kernel; BAR, Bartlett kernel.

Specifically, we denote by $g(x)$ the natural cubic spline passing through points $(0, 0)$, $(0.33, 0.2)$, $(0.66, 0.8)$, $(1, 1)$ and we generate a random sample $\{\xi_t\}_{t=1}^n$ from a uniform distribution over $(-1, 1)$. For $x \in \mathcal{S} := [0, 1]$, we define $h(x) = \frac{1}{2}\{(1 - \xi_t)g(x) + (1 + \xi_t)x\}$ and $\varepsilon_t(x) = \frac{1}{2}[(1 + \xi_t)g\{h^{-1}(x)\} + (1 - \xi_t)h^{-1}(x)]$. Then we generate a sequence of quantile functions $\{T_t\}_{t=1}^n$ by

- (i) ATM(0): $T_t = \varepsilon_t$;
- (ii) ATM(1): $T_t = \beta_1 \odot T_{t-1} \oplus \varepsilon_t$;
- (iii) $T_t = \beta_4 \odot T_{t-4} \oplus \beta_3 \odot T_{t-3} \oplus \beta_2 \odot T_{t-2} \oplus \beta_1 \odot T_{t-1} \oplus \varepsilon_t$, where $(T_1 \oplus T_2)(x) = T_2 \circ T_1(x) = T_2\{T_1(x)\}$ and, for $\beta \in [-1, 1]$,

$$\beta \odot T(x) = \begin{cases} x + \beta\{T(x) - x\}, & 0 < \beta \leq 1, \\ x, & \beta = 0, \\ x + \beta\{x - T^{-1}(x)\}, & -1 \leq \beta < 0. \end{cases}$$

For ATM(1), we consider $\beta_1 = 0.5\rho$ with $\rho \in \{0.2, 0.5, 1\}$, whereas for ATM(4), we consider $\beta = \rho(0.2, -0.5, 0.1, -0.3)^T$ with $\rho \in \{0.5, 0.8, 1\}$. Based on the simulated quantile functions $\{T_t(x) : x \in \mathcal{S}\}_{t=1}^n$, we generate $\{Y_t(x) : x \in [0, 1]\}_{t=1}^n$ by $Y_t(x) = T_t^{-1}(x)$.

For the metric space Ω , we consider several metrics to evaluate the distance between two distributions. In particular, we use F, G to denote two cumulative distribution functions, use F^{-1}, G^{-1} to denote the corresponding quantile functions and use f, g to denote the corresponding density functions. Then we have the following metrics.

- (i) 1-Wasserstein (W1) metric: $d_{W1}(F, G) = \int_0^1 |F^{-1}(t) - G^{-1}(t)| dt$.
- (ii) 2-Wasserstein (W2) metric: $d_{W2}(F, G) = [\int_0^1 \{F^{-1}(t) - G^{-1}(t)\}^2 dt]^{1/2}$.
- (iii) Kolmogorov–Smirnov (KS) metric: $d_{KS}(F, G) = \sup_{x \in \mathcal{S}} |F(x) - G(x)|$.
- (iv) Kullback–Leibler (KL) metric:

$$d_{KL}(F, G) = \frac{1}{2} \left[\int_{x \in \mathcal{S}} \log\{F(dx)/G(dx)\} F(dx) + \int_{x \in \mathcal{S}} \log\{G(dx)/F(dx)\} G(dx) \right].$$

- (v) Itakura–Saito (IS) metric:

$$d_{IS}(F, G) = (2|\mathcal{S}|)^{-1} \int_{x \in \mathcal{S}} \left[\left\{ \frac{f(x)}{g(x)} - \log \left(\frac{f(x)}{g(x)} \right) - 1 \right\} + \left\{ \frac{g(x)}{f(x)} - \log \left(\frac{g(x)}{f(x)} \right) - 1 \right\} \right] dx.$$

- (vi) Log-spectral (LS) metric: $d_{LS}(F, G) = (|\mathcal{S}|^{-1} \int_{x \in \mathcal{S}} [10 \log_{10}\{f(x)/g(x)\}]^2 dx)^{1/2}$.

In practice, we compute the Wasserstein distances directly using the simulated quantile functions $\{T_t(x)\}_{t=1}^n$, and compute all the other distances using the R package *seewave* (R Development Core Team, 2024) with the probability functions estimated from the simulated cumulative distribution functions. To the best of our knowledge, there is no existing literature for testing the serial dependence within a sequence of univariate distributions; hence we only present the empirical rejection rates of the proposed method in Table 3. All the numerical results are based on 2500 Monte Carlo replicates.

The overall pattern shown in Table 3 generally matches those observed in § 5.1 and § 5.2. Specifically, the type of test statistic and the method used to approximate the critical value have little influence on both the size and the power. Among all the metrics examined in Table 3, we observe no significant disparities in terms of the size accuracy, whereas the Itakura–Saito metric and the Kullback–Leibler metric exhibit the dominating power in most cases.

6. APPLICATIONS

6.1. Cumulative intraday returns

Modelling financial returns using functional data approaches has attracted much attention in the literature; see, e.g., Aue et al. (2017), Shang (2017) and Cerovecki et al. (2019). In the first application, we apply the proposed tests to the cumulative intraday returns of two stocks in 2007: IBM and McDonald's Corporation. Following Gabrys et al. (2010), we define the cumulative intraday returns as

$$X_t(s) = 100[\log\{P_t(s)/P_t(0)\}], \quad s \in [0, 1],$$

Table 3. Empirical rejection rates (%) of univariate distributions when $n = 200$

		W1		W2		KS		KL		IS		LS	
		Permt	Boot	Permt	Boot	Permt	Boot	Permt	Boot	Permt	Boot	Permt	Boot
ATM(0)	CvM	4.9	5.2	4.9	5.0	5.0	5.0	5.0	5.0	4.7	4.5	5.0	4.9
	KS	4.9	5.2	5.0	5.1	5.0	5.7	4.9	5.0	4.9	5.0	5.3	4.9
ATM(1) $\rho = 0.2$	CvM	18.6	18.3	17.8	18.0	17.8	17.3	23.9	23.4	27.2	26.2	20.3	20.1
	KS	17.2	17.6	17.5	17.1	16.4	16.4	22.4	22.8	26.4	25.3	18.8	18.6
$\rho = 0.5$	CvM	62.4	62.3	62.2	62.4	60.3	59.5	74.5	73.2	81.4	78.9	69.6	68.9
	KS	60.7	60.4	59.9	60.3	57.9	57.5	72.8	71.6	80.0	78.2	67.8	67.6
$\rho = 1.0$	CvM	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	97.0	97.8	100.0	100.0
	KS	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	97.0	97.7	100.0	100.0
ATM(4) $\rho = 0.5$	CvM	17.6	18.0	17.8	17.6	17.3	17.1	24.0	24.5	27.9	26.5	18.6	18.6
	KS	18.4	18.8	18.8	18.6	17.5	18.0	25.3	25.3	28.9	28.0	20.2	19.8
$\rho = 0.8$	CvM	73.0	73.2	73.2	73.7	71.7	71.4	84.6	83.4	86.2	82.4	76.2	75.9
	KS	73.5	73.6	74.2	74.5	72.4	71.8	85.5	84.0	87.0	83.8	75.6	75.0
$\rho = 1.0$	CvM	99.6	99.6	99.7	99.7	99.5	99.5	99.8	99.8	99.6	98.8	99.8	99.7
	KS	99.4	99.6	99.6	99.6	99.5	99.6	99.8	99.9	99.7	98.9	99.8	99.7

Table 4. The p -values (%) for white noise testing in cumulative intraday returns of IBM and MacDonald's stock in 2007

Test	CvM	KS	GK1	GK3	GK5	HHHR5	HHHR10	HHHR30	HHHR50	Zhang1	Zhang5	Zhang10
IBM	0.8	1.0	0.3	0.8	0.4	0.0	0.0	0.2	10.0	4.0	10.0	2.7
McDonald	85.8	93.8	41.6	4.4	9.8	78.5	20.7	42.4	15.3	55.2	30.1	34.4

where $P_t(s)$ denotes the stock price at rescaled trading period s and $P_t(0)$ denotes the market opening price. The daily trading data are collected over a one-minute frequency with a total 390 data points per day. The data are then smoothed using 20 Fourier basis functions. We consider the proposed Cramér–von Mises test and Kolmogorov–Smirnov test, along with other competing methods, including GK at lag $h = 1, 3, 5$, HHHR with truncation lag $h = 5, 10, 30, 50$ and Zhang with $b = 1, 5, 10$. The p -values of all tests are summarized in Table 4. From Table 4, we find that all tests except for HHHR50 reject H_0 at the 5% level for the IBM stock, and all except for GK3 fail to reject H_0 for MacDonald's stock. This finding is consistent with Zhang (2016) using five-minute data and cubic B -spline basis functions. Here, the constructed price process could be contaminated by microstructure noise (Aït-Sahalia & Yu, 2009) and we conjecture that its presence could compromise power due to a diminished signal-to-noise ratio. We leave this for future research.

6.2. Mortality data

The Human Mortality Database (<https://www.mortality.org/Home/Index>) provides scientific researchers with high-quality harmonized mortality and population estimates. For example, Fig. 1 gives a visual demonstration of the female mortality distribution time series in the UK from 1970 to 2014 for women aged between 20 and 105.

The yearly age-at-death distribution for a given country can be naturally viewed as a random element in the space of univariate distributions, which has been analysed in many studies regarding modelling aspects for non-Euclidean-valued random objects; see, e.g., Petersen & Müller (2019) and Dubey & Müller (2020). However, many of them implicitly assume temporal independence for the sequence of random distributions, which could be doubtful. We now formally test for this assumption.

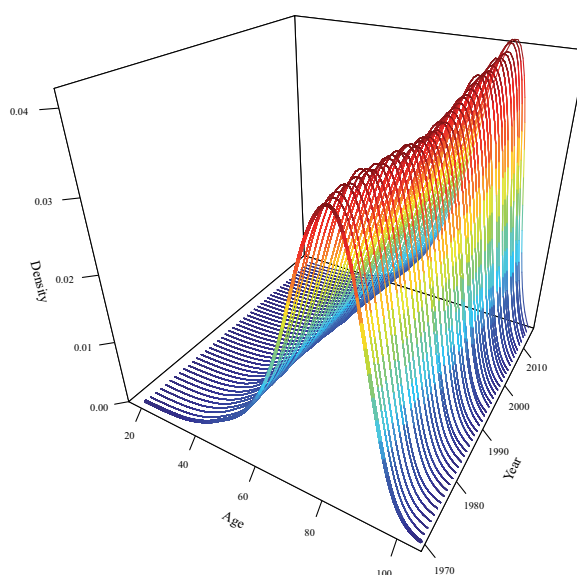


Fig. 1. UK female yearly age-at-death distribution, 1970–2014.

We analyse the yearly female mortality data in 27 European developed countries from 1970 to 2014. After applying our proposed Cramér–von Mises test and Kolmogorov–Smirnov test with Wasserstein-1 and Wasserstein-2 metrics, we find that all tests for all the countries reject H_0 at the 5% level. This provides strong evidence of temporal dependence in a mortality distribution time series. Therefore, more caution should be taken when it comes to the modelling of mortality data, as temporal dependence may not be negligible.

7. DISCUSSION

This paper is concerned with serial dependence testing of object-valued time series, which is typically conducted before any specific model is fitted. There is a recent literature on modelling temporal dependence in object-valued time series, e.g., distributional autoregression in [Zhang et al. \(2022\)](#), [Zhu & Müller \(2023a\)](#) and [Ghodrati & Panaretos \(2023\)](#), and spherical autoregression in [Zhu & Müller \(2023b\)](#), among others. It would be interesting to investigate how to extend the test in this paper to model diagnostic checking. This requires researchers to properly define model residuals, which is a challenging task for object-valued data in metric spaces. In addition, we assume that the object-valued data are fully observed in the paper, whereas in real applications they are usually generated by pre-processing Euclidean-valued data. In the realm of Hilbert space, [Zhang & Wang \(2016\)](#) demonstrated that presmoothing in functional data analysis can exhibit a phase transition phenomenon, depending on the resolution of discretized observations. It would also be helpful to investigate how pre-processing would impact the behaviour of our test. Finally, we assume that the distance metric $d(\cdot, \cdot)$ is given a priori, but its choice can play a significant role in data analysis; it usually depends on the underlying inference problem and the characteristics of the dataset. We briefly discuss this issue in the [Supplementary Material](#), but the theoretical investigation, such as that of optimality, is beyond the scope of this paper. We leave these topics for future research.

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SUPPLEMENTARY MATERIAL

The [Supplementary Material](#) contains additional numerical studies, auxiliary lemmas and all the technical details and proofs. The code is available at <https://github.com/hjgao117/JiangGaoShao>.

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